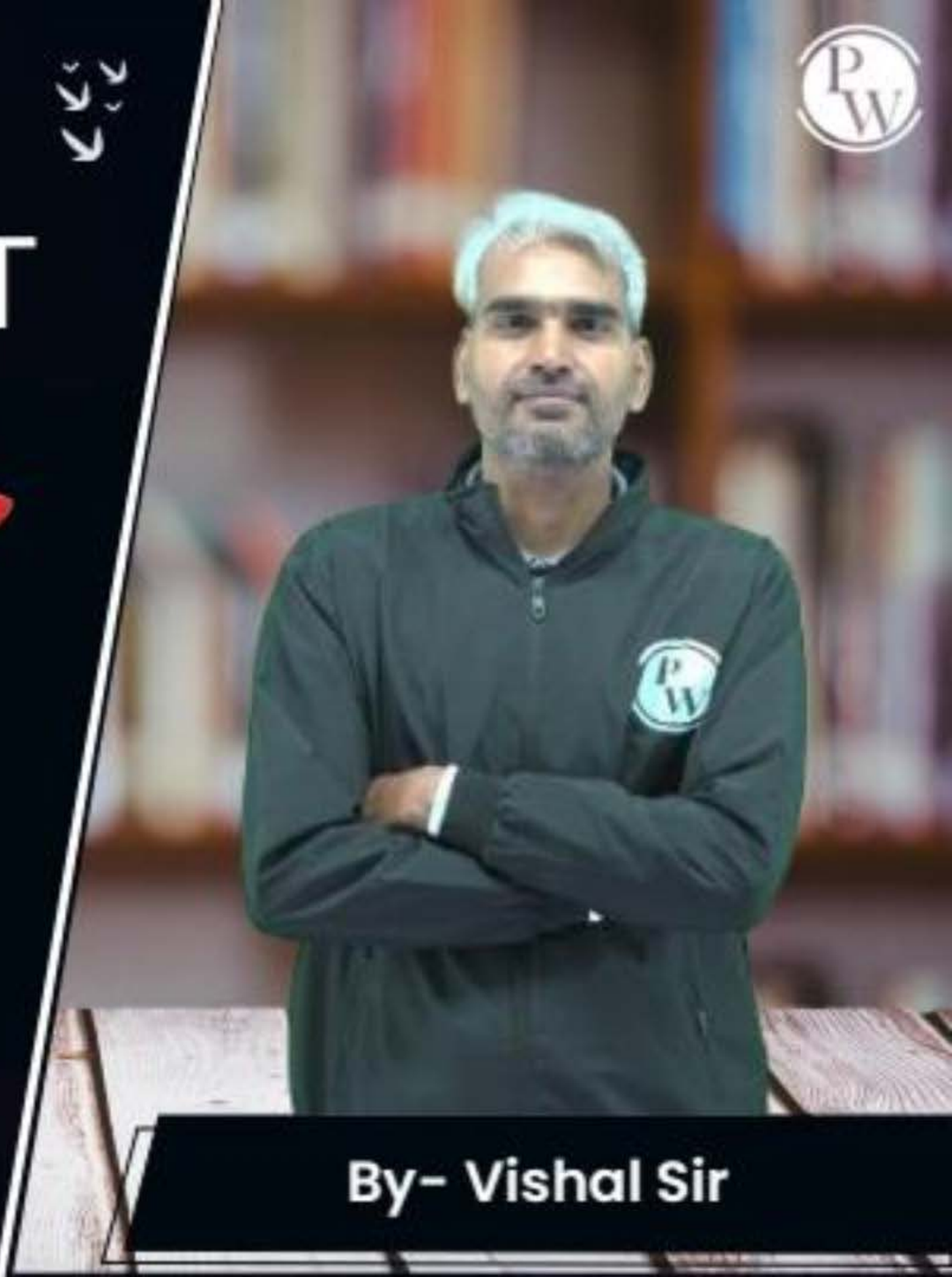
Computer Science & IT



Database Management System

Query Languages

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



- Topic: SQL
- Topic: SQL commands
- Topic: SQL clauses
- Topic: Aggregate functions

Slide

Topics to be Covered



- Topic: SQL commands
- Topic: SQL clauses
- Topic: Order of execution

Slide



Topic : Commands in SQL

- CREATE TABLE - creates a new table
- ALTER TABLE - modifies a table
- DROP TABLE - deletes a table

} Data definition language

- INSERT - inserts new data into a database
- DELETE - deletes data from a database
- UPDATE - updates data in a database

} Data Manipulation language

- SELECT - extracts data from a database

} Data query language



Topic : Commands in SQL

- CREATE TABLE - creates a new table

Name of the table

```
CREATE TABLE Student (
        Sid varchar(25),
        Sname varchar(25),
        Marks int,
        );
```

Names of attributes

Data types

STUDENT

Sid	Sname	Marks



Topic : Commands in SQL

- ALTER TABLE - modifies a table

```
ALTER TABLE Student
ADD Branch varchar(25);
```

DROP
↳ to delete existing attribute.

STUDENT

Sid	Sname	Marks	Branch



Topic : Commands in SQL

- DROP TABLE - deletes a table

DROP TABLE *Student*;

{ It will delete the complete Student table from database }

- TRUNCATE TABLE - deletes complete data from the table without deleting the structure of the table

TRUNCATE TABLE Student;



Topic : Commands in SQL

- INSERT - inserts new data into a database

- INSERT INTO *Student* VALUES (*S1*, *A*, *35*, CS);

STUDENT

Sid	Sname	Marks	Branch
S1	A	35	CS



Topic : Commands in SQL

- DELETE - deletes data from a database

tuple



Topic : Commands in SQL

- UPDATE - updates data in a database

```
UPDATE Student
SET Marks = Marks + 5;
```

STUDENT

Sid	Sname	Marks	Branch
S1	A	35+5=40	CS



Topic : SQL clauses

- FROM - The FROM clause in SQL is used to select the database tables
- WHERE - The WHERE clause in SQL is used to retrieve the data from the database based on conditions specified with WHERE clause.
- GROUP BY - GROUP BY clause is used to group the result of WHERE clause.
- HAVING - HAVING clause can be used with GROUP BY clause. It is used to specify a search condition for a group in the database tables.
- ORDER BY - The ORDER BY clause in SQL is used for sorting the records of the database



Topic : SQL clauses

Select $A_1, A_2, \ldots A_n$
From $R_1, R_2, \ldots R_m$
Where (condn)
Group by (attribute)
Having (condn)

(i) FROM:- From clause is used to select the tables from the database.

Select * ← Every thing (All attributes)
From Student

It will produce Complete Student table as output.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

Select $A_1, A_2, \ldots A_n$
From $R_1, R_2, \ldots R_m$
Where (condn)
Group by (attribute)
Having (condn)

(i) FROM:- From clause is used to select the tables from the database.

Select Sname
from Student ⇒

Sname
A
A
B
A
C
C

without distinct keyword, duplicate tuples may be present in output

Select Distinct Sname
From Student ⇒

Sname
A
B
C

Note ⩵ Select distinct ≡ Projection (Π) of Relational Algebra

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

From Student, Enroll ≡ Student X Enroll
in Relational Algebra

Equivalent to
Cross product
in Rel^{l} Algebra



Topic : SQL clauses

(ii) WHERE:- Used to retrieve the data from the database based on conditions specified with WHERE clause.

```
Select *
   From Student
      Where (Marks ≥ 40)
```

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

(ii) WHERE:- Used to retrieve the data from the database based on conditions specified with WHERE clause.

```
Select  Sid
  From  Student
    Where (marks = 40)
```

⇒

Sid
S1
S5

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

(ii) WHERE:- Used to retrieve the data from the database based on conditions specified with WHERE clause.

```
Select  Sid
   From  Student
      Where  Marks  between  10  and  40 ;
```

"inclusive" i.e. both 10 & 40 are in Range



STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC



Topic : SQL clauses

(iii) GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

If Where clause is not present then Group by clause will be applied on Complete table.

Query: Retrieve names of all branches along with maximum marks in that branch.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

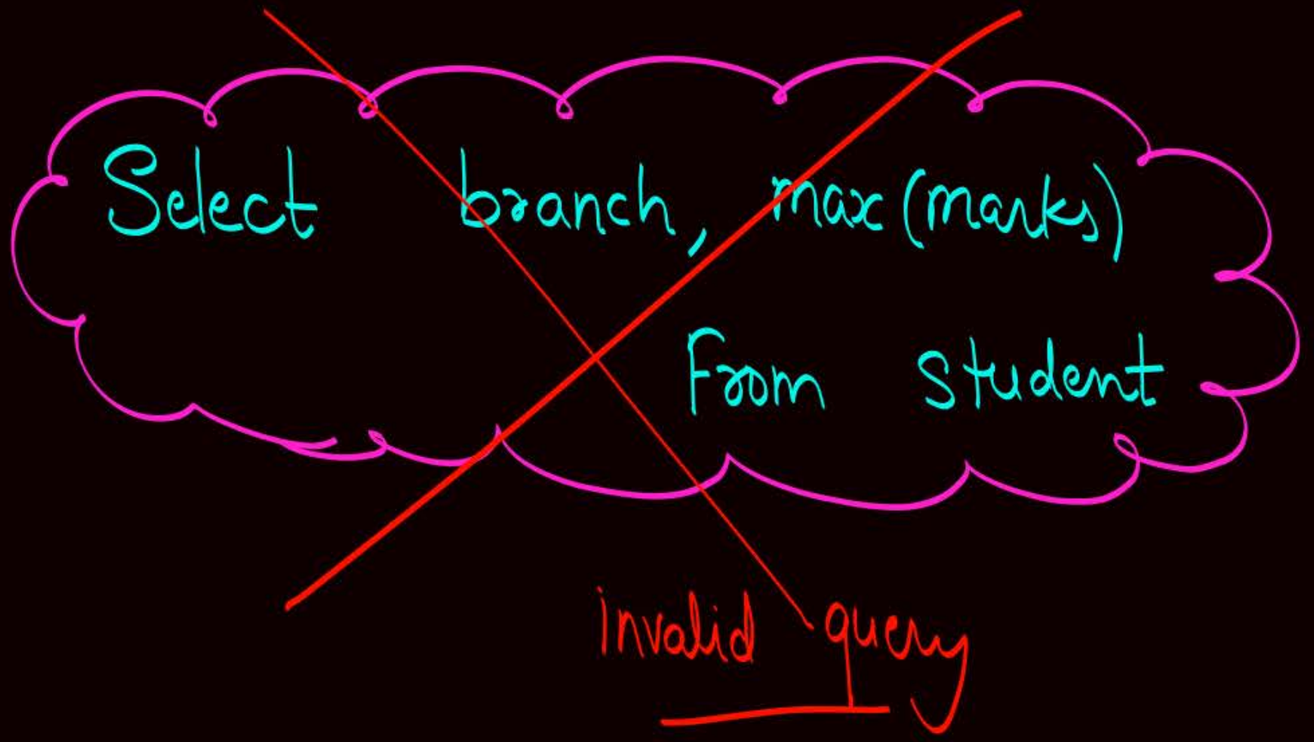
Query :- Select maximum marks from Student table.

Soln

```
Select max(marks)
From Student
```

⇒ Ans = 60

Query: Retrieve names of all branches along with Maximum marks from that branch.



We can not select branch attribute along with aggregate function max(marks) because attribute branch is not present in group by clause



Topic : SQL clauses

NOTE:-

1. We can not select any attribute in SELECT clause along with aggregate function until those attributes are present in GROUP BY clause.

2. If aggregate function is used along with GROUP BY clause, then aggregate function is applied on each group.

If Group by clause is not present in the query then, aggregate function will be applied on the output of where clause



Topic : SQL clauses

GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

Query: Retrieve names of all branches along with maximum marks in that branch.

Select Branch, max(marks)
From Student
Group by (Branch)

Where clause is not present ∴ Groupby will be applied on complete table.

=>

CS	60
IT	40
EC	60

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

60
40
60



Topic : SQL clauses

(iv) HAVING:- HAVING condition is applied on each group. { Provided group by Clause is present }

Query: Retrieve branch names with average marks more than or equal to 40.

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

CS: 40 ≥ 40

IT: 30 ≠ 40

EC: 60 ≥ 40

```
Select Branch
From Student
Group by (Branch)
Having (Avg(Marks) ≥ 40)
```

Group for which Having Condⁿ is true will be selected

=

Branch
CS
EC



Topic : SQL clauses

NOTE:-

1. WHERE condition is applied on each tuple whereas HAVING condition is applied on each group.
2. We can use HAVING condition without GROUP BY clause, but in that case HAVING condition will be applied on each tuple. i.e., without GROUP BY clause HAVING clause will degenerate into WHERE clause.



Topic : SQL clauses

(v) ORDER BY:- This clause is used to sort the result in ascending or descending order based on values of attribute specified with ORDER BY clause.

By default order is ascending order.

ASC or DESC

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

```
Select Sid, Sname
    From Student
        Where (marks = 40)
            Order by (Sid)
```

⇒

Sid	Sname
S_1	A
S_5	C

↑ by default Ascending order i.e: (ASC)

```
Select Sid, Sname
    From Student
        Where (marks = 40)
            Order by (Sid) DESC
```

↑ for descending order

Sid	Sname
S_5	C
S_1	A

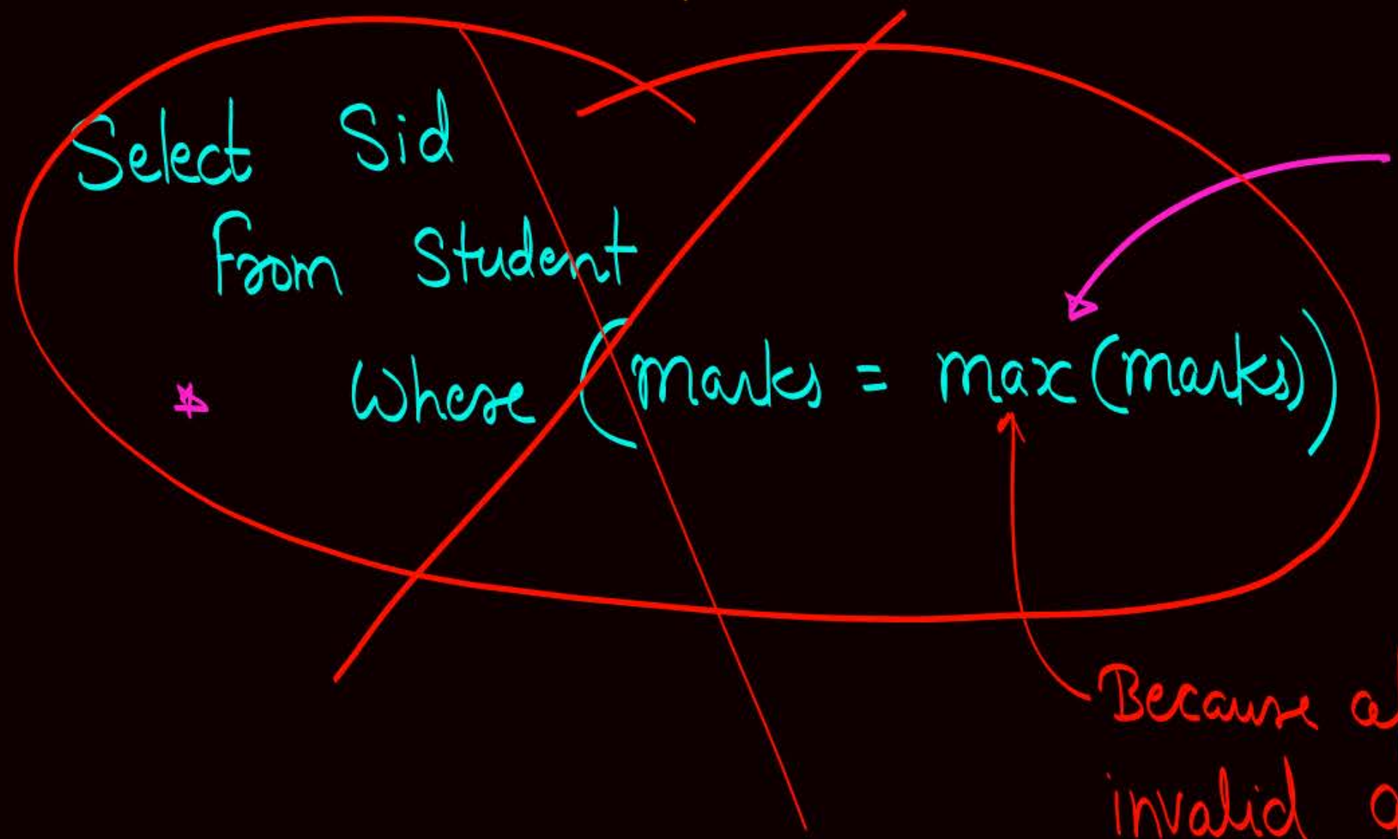


Topic : Order of execution

Order of Execution:-

1. From
2. Where
3. Group By
4. Having
5. Select
6. Order BY

Q: Retrieve Sids of the students with maximum marks.



Q: Retrieve Sids of the students with maximum marks.

```
Select Sid
From Student
Where (marks = (Select Max(Marks) From Student))
```

Sub-query (or) Nested query

Main query (outer query)

(inner query)

Result = 60

inner query will be first

then

outer query will be executed

O/P =

Sid
S3
S4

Tomorrow's Topic

Nested queries



Most important topics of DBMS



2 mins Summary

- **Topic** SQL commands
- **Topic** SQL clauses
- **Topic** Order of execution

Slide

THANK - YOU